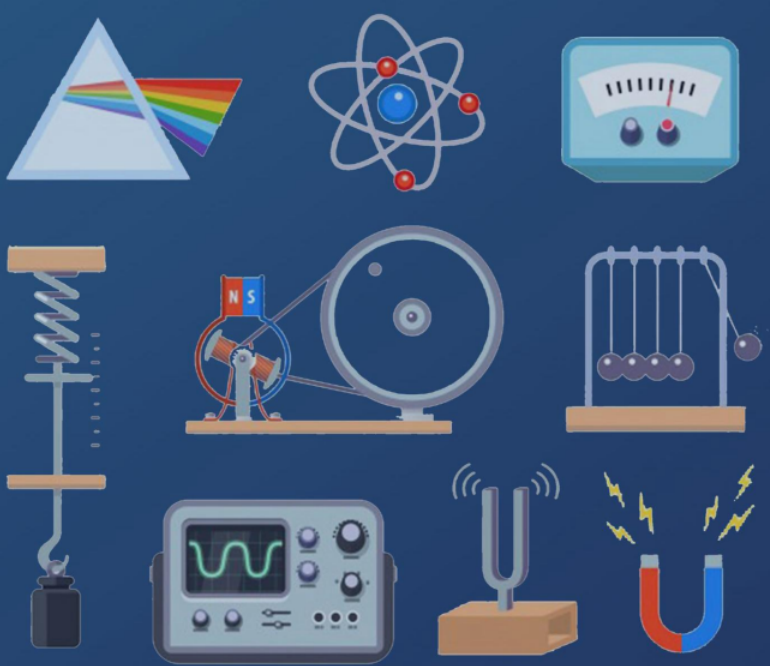
PHYSICS

Circular Motion



SHEET

eSaral.com

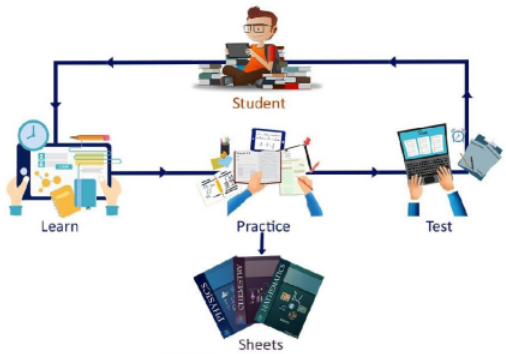
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 5
5.	Exercise 2	Page - 7
6.	Exercise 2A	Page - 9
7.	Answer Key	Page - 12

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Angular Variables	1,2,3			
2	Kinematics of Uniform Circular Motion	4,5	1		
3	Kinematics of Non-Uniform Circular Motion	6,7	2,4	1,2,3,4	1,4
4	Curvilinear Motion	8		5	2,6,7
5	Dynamics of Circular Motion	9,10,11,12	5,6,7,8,9	6,7,8	5,8
6	Banking of Roads		3	9	9,10
7	Centrifugal Force	13,14,15		10	
8	Angular Velocity in General	16,17,18	10		3

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Angular Variables

Q1. A wheel is of diameter 1m. If it makes 30 revolution per second, then the linear speed of a point on its circumference will be :-

(A) 30 π m/s (B) π m/s

(C) 60 π m/s (D) $\frac{\pi}{2}$ m/s

Q2. A wheel is rotating at 900 rpm about its axis when the power is cut off. It comes to rest in one minute. The angular retardation (assuming it to be uniform) in radian/sec^2 is:

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{8}$

Q3. A wheel initiallly at rest starts moving about its axis with constant angular acceleration. In first 2sec. it rotates through angle θ_1 and in further 2sec. it rotates through angle θ_2. The ratio θ_2/θ_1 is :-

(A) 1 (B) 2

(C) 3 (D) 5

Kinematics of Uniform Circular Motion

Q4. In uniform circular motion :-

(A) Velocity is constant

(B) Acceleration is constant

(C) Speed is constant

(D) Linear momentum is constant

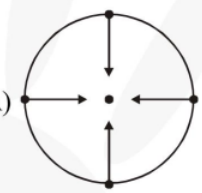
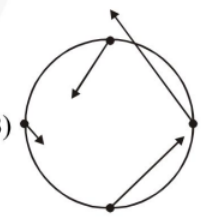
Q5. Two cars of masses m_1 & m_2 are moving along the circular paths of radius r_1 & r_2 respectively. Their speeds are such that they complete one round in same time. The ratio of angular speeds of two cars is :-

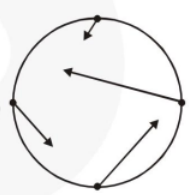
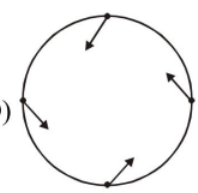
(A) $m_1 : m_2$ (B) $r_1 : r_2$

(C) 1 : 1 (D) $m_1r_1 : m_2r_2$

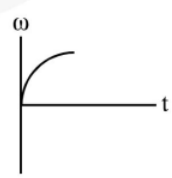
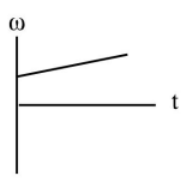
Kinematics of Non-Uniform Circular Motion

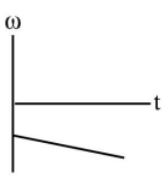
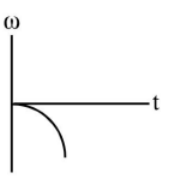
Q6. A car speeds up with constant magnitude of tangential acceleration in circular path moving in anticlockwise direction. Which of the following figure represents acceleration of the car?

(A)  (B) 

(C)  (D) 

Q7. The graphs below show angular velocity as a function of time. In which one is the magnitude of the angular acceleration constantly decreasing?

(A)  (B) 

(C)  (D) 

Curvilinear Motion

Q8. A body is projected with a velocity 10 ms^{-1} at an inclination 45° to the horizontal. Minimum radius of curvature of the trajectory described by the particle is _____.

(A) 5m (B) 10 m

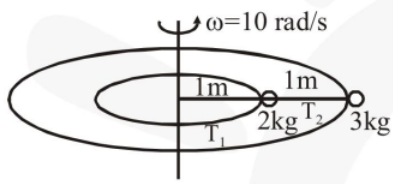
(C) 0 m (D) 2.5 m

Dynamics of Circular Motion

Q9. A stone of mass 1 kg, tied to the end of a string of length 1 m, is whirled in a horizontal circle with a uniform angular velocity of 2 rad/s. The tension in the string is (in N) :

(A) 2 (B) 1/2 (C) 4 (D) 1/4

Q10. Two particles of mass 2 kg and 3 kg connected to a rope of length 2m as shown and the system is revolved in horizontal circular path with angular velocity of 10 rad/s then what will be the value of tension T_1 and T_2 :-



(A) 200 N, 600 N (B) 600 N, 200 N

(C) 800 N, 600 N (D) 600 N, 800 N

Q11. What is the minimum radius of a circle along which a cyclist can ride with a velocity 18 km/hr if the coefficient of friction between the tyres and the road is $\mu = 0.5$

(take $g = 10\ m/s^2$).

(A) 5 m (B) 10 m

(C) 2 m (D) None of these

Q12. A mass is revolving in a circle which lies in a plane of paper. The direction of angular acceleration can be :-

(A) perpendicular to radius and velocity

(B) towards the radius

(C) tangential

(D) at right angle to angular velocity

Centrifugal Force

Q13. A man is standing on a rough ($\mu = 0.5$) horizontal disc rotating with constant angular velocity of 5 rad/sec. At what distance from centre should he stand so that he does not slip on the disc?

(A) $R \le 0.2$ m (B) $R > 0.2$ m

(C) $R > 0.5$ m (D) $R > 0.3$ m

Q14. A particle of mass m is tied to a light string and rotated with a speed v along a circular path of radius r. If T = tension in the string and mg = gravitational force on the particle then the actual forces acting on the particle are :

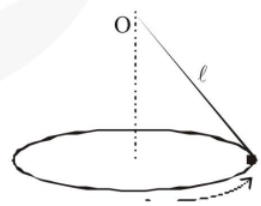
(A) mg and T only

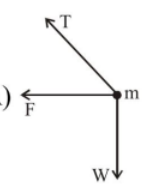
(B) mg, T and an additional force of $\frac{mv^2}{r}$ directed inwards.

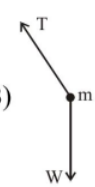
(C) mg, T and an additional force of $\frac{mv^2}{r}$ directed outwards.

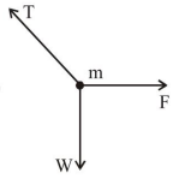
(D) only a force $\frac{mv^2}{r}$ directed outwards

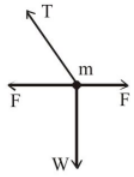
Q15. A point mass m is suspended from a light thread of length ℓ, fixed at O, is whirled in a horizontal circle at constant speed as shown. From your point of view, stationary with respect to the mass, the forces on the mass are :



(A) 

(B) 

(C) 

(D) 

Angular Velocity in General

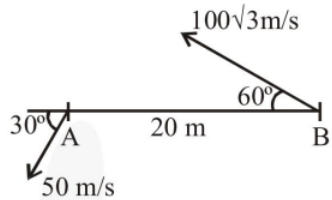
Q16. A body is executing circular motion in the vertical plane containing directions. If the direction of velocity $(\vec{v})$ at the top most point is towards west, what is the direction of angular velocity $(\vec{\omega})$?

(A) east (B) west
(C) north (D) south

Q17. If the magnitude of velocity in the previous question is decreasing with time, what is the direction of angular acceleration $(\vec{\alpha})$?

(A) east (B) west (C) north (D) south

Q18. Velocity of two particles A and B separated by distance 20 m apart is as shown in figure then what will be the angular velocity of B with respect to A



(A) 4.75 rad/s (B) 8.75 rad/s
(C) 16.75 rad/s (D) None of these

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following statements is/are TRUE for a particle moving in a circle with a constant angular speed ?

(A) The velocity vector is tangent to the circle
(B) The acceleration vector is tangent to the circle
(C) The acceleration vector points to the centre of the circle
(D) The velocity and acceleration vectors are perpendicular to each other

Q2. A particle is performing non uniform circular motion. Choose the CORRECT statement(s) :-

(A) The particle moves with variable speed
(B) The acceleration is always normal to velocity
(C) The particle moves with variable acceleration
(D) Net Acceleration and velocity are parallel to each other always.

Q3. For a curved track of radius R, banked at angle θ (Take $v_0 = \sqrt{Rg\tan\theta}$).

(A) a vehicle moving with a speed v_0 is able to negotiate the curve without calling friction into play at all
(B) a vehicle moving with any speed $v > v_0$ is always able to negotiate the curve, with friction called into play
(C) a vehicle moving with any speed $v < v_0$ must have the force of friction into play
(D) the minimum value of the angle of banking for a vehicle parked on the banked road can stay there without slipping, is given by $\theta = \tan^{-1}\mu_0$ (μ_0 = coefficient of static friction)

MATCHING LIST TYPE

Each question has THREE(03) ***matching lists : LIST I , LIST II and LIST III*** *FOUR options are given representing matching of elements from LIST I , LIST II and LIST III ONLY ONE of these four options corresponds to a correct matching.For each question, choose the option corresponding to the correct matching.*

Q4. A particle move in a circular path with varying $\theta = 3t^3 - 5t^2 + 4$, then -

	TIME	ANGULAR VELOCITY	ANGULAR ACCELERATION
(1)	t = 2	(P) 0 rad/sec	(L) 26 rad/sec^2
(2)	t = 10/9	(Q) 16 rad/sec	(M) 10 rad/sec^2
(3)	t = 0	(R) 51 rad/sec	(N) – 10 rad/sec^2
(4)	t = 3	(S) 14 rad/sec	(O) 44 rad/sec^2

(A) (1) → Q, L; (2) → P, M;(3) → P, N; (4) → R,O
(B) (1) → Q, N; (2) → Q, M;(3) → P, N; (4) → R,O
(C) (1) → Q, L; (2) → P, M;(3) → R, N; (4) → R,O
(D) (1) → S, L; (2) → P, M;(3) → P, N; (4) → R,O

PARAGRAPH TYPE

This section contains ***PARAGRAPHS****. Based on each paragraph, there are questions. Each question has FOUR options.* ***ONLY ONE*** *of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

A block is rotating in a circular track and angular speed of the block is given $\omega = at^2$, where ω is in rad/s and t is in seconds, a is a constant and equal to 2 rad/s^3. Mass of the block is 1 kg. Radius of circular track is 1 m.

Q5. Net force acting on the block at t = 2 s is -

(A) 8 N (B) $8\sqrt{65}$ N
(C) 32 N (D) 4 N

Q6. Angle between velocity vector and acceleration vector is -

(A) less than 90°
(B) greater than 90°
(C) equal to 90°
(D) data is insufficient

Q7. Time at which rate of change of speed is equal to $\frac{1}{\sqrt{2}}$ times of net acceleration -

(A) $\frac{1}{2}$s (B) 1s
(C) 2s (D) 4s

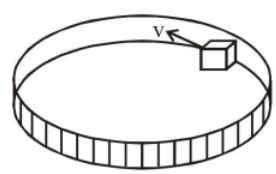
SUBJECTIVE TYPE

Q8. A block of mass m moves with speed v against a smooth, fixed vertical circular groove of radius r kept on smooth horizontal surface.

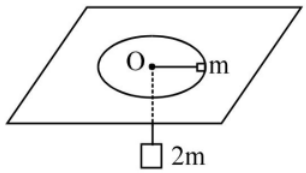
Find :

(i) normal reaction of the floor on the block.

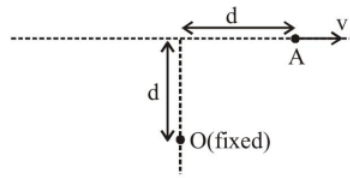
(ii) normal reaction of the vertical wall on the block.



Q9. A mass m rotating freely in a horizontal circle of radius 1 m on a frictionless smooth table supports a stationary mass 2m, attached to the other end of the string passing through smooth hole O in table, hanging vertically. Find the angular velocity of rotation.



Q10. Find angular velocity of A with respect to O at the instant shown in the figure.



Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If angular velocity of a disc depends an angle rotated θ as $\omega = \theta^2 + 2\theta$, then its angular acceleration at $\theta = 1$ rad is :-

(A) 8 rad/sec^2 (B) 10 rad/sec^2

(C) 12 rad/sec^2 (D) None

Q2. A car is moving on a circular path of radius 500m with a speed of 30 m/s at some instant. If its speed increases at the rate of 2m/s^2, its resultant acceleration at that instant is :-

(A) 2.5 m/s^2 (B) 2.7 m/s^2

(C) 7.2 m/s^2 (D) 4.0 m/s^2

Q3. A particle is moving on a circular path with a velocity $v = 1.5\ t^2 + 2t$, where v is in m/s and t in seconds. If radius of circular path is 2cm, then angular acceleration of the particle at $t = 2$s is :-

(A) 4 rad/s^2 (B) 40 rad/s^2

(C) 400 rad/s^2 (D) 0.4 rad/s^2

Q4. Let $\vec{v}(t)$ be the velocity of a particle at time t. Then INCORRECT statement is:

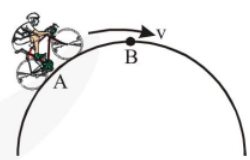
(A) $|d\vec{v}(t) / dt|$ and $d|\vec{v}(t)| / dt$ are always equal

(B) $|d\vec{v}(t) / dt|$ and $d|\vec{v}(t)| / dt$ may be equal

(C) $d|\vec{v}(t)| / dt$ can be zero while $|d\vec{v}(t) / dt|$ is not zero

(D) $d|\vec{v}(t)| / dt \neq 0$ implies $|d\vec{v}(t) / dt| \neq 0$

Q5. A motorcylist is going on an overbridge of radius r maintaining a constant speed v. As the motorcyclist goes up on the overbridge, the normal force :-



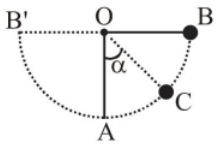
(A) Increases ($N_B > N_A$)

(B) Decreases ($N_B < N_A$)

(C) Remains same ($N_B = N_A$)

(D) Nothing can be said

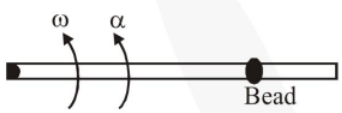
Q6. A simple pendulum is vibrating with an angular amplitude of 90° as shown in the adjoining figure. For what value of α, is the acceleration directed horizontally :-



(A) 0° (B) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(C) 90° (D) $\cos^{-1}\left(\frac{2}{3}\right)$

Q7. A bead is constrained to move on rod in gravity free space as shown in figure. The rod is rotating with angular velocity ω and angular acceleration α about its end. If μ is coefficient of friction. Mark the correct option. (Rod rotates in the plane of paper.)



(A) If $\mu = \frac{\omega^2}{\alpha}$ friction on bead is static in nature

(B) If $\mu \geq \frac{\omega^2}{\alpha}$ friction on bead is kinetic in nature

(C) If $\mu \leq \frac{\omega^2}{\alpha}$ friction is static

(D) If bead does not slide relative to rod. Friction will not exist between bead and rod.

Q8. A ring of radius r and mass per unit length m rotates with an angular velocity ω in free space. The tension in the ring is :

(A) zero (B) $\frac{1}{2}m\omega^2r^2$

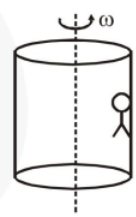
(C) $m\omega^2r^2$ (D) $mr\omega^2$

Q9. A circular road of radius 1000 m has banking angle 45°. The maximum safe speed of a car having a mass 2000 kg will be nearly, if the coefficient of friction between tyre and road is 0.5 :-

(A) 172 m/s (B) 124 m/s

(C) 99 m/s (D) 86 m/s

Q10. In a rotar a hollow vertical cylindrical structure rotates about its axis and a person rests against the inner wall. At a perticular speed of the rotar, the floor below the person is removed and the person hangs resting against the wall without any floor. If radius of rotar is 2m and $\mu_s = 0.2$ what will be the minimum angular velocity at which floor may be removed :-



(A) 2 rad/s (B) 5 rad/s

(C) 10 rad/s (D) None of them

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A particle is moving in a circular path and its acceleration vector is making an angle of 30° with the velocity vector and the magnitude of acceleration is 2 m/s^2, then -

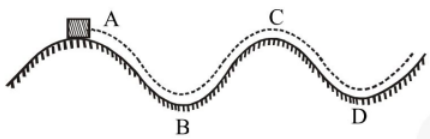
(A) Centripetal acceleration of particle 1 m/s^2

(B) Centripetal acceleration of particle $\sqrt{3}$ m/s^2

(C) Tangential acceleration of particle $\sqrt{3}$ m/s^2

(D) Tangential acceleration of particle 1 m/s^2

Q2. A body is moving with constant velocity on the path shown.



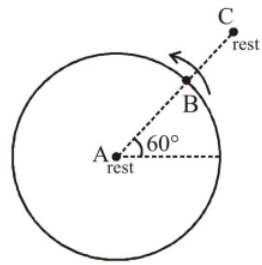
(A) Normal reaction exerted on body by path is greatest at N_B

(B) Normal reaction exerted on body by path is least at N_C

(C) Normal force is greater than weight at B

(D) Normal force is lesser than weight at A, D

Q3. Diagram shows three particles A, B & C. Particle B is moving in a circle of radius R with constant angular velocity ω anti-clockwise, A & C are at rest, AB = BC = R. Choose the correct statement(s).



(A) Angular velocity of B wrt A is ω clockwise

(B) Angular velocity of B wrt A is ω anticlockwise

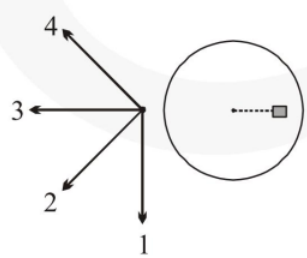
(C) Angular velocity of B wrt C is constant & equal to ω

(D) Angular velocity of B wrt C is not constant and is ≤ ω

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q4. A block is placed on a horizontal table which can rotate about its axis.The block is placed at a certain distance from centre as shown in figure. Table rotates such that particle does not slide. Select possible direction of net acceleration of block at the instant shown in figure.



	Column-I		Column-II
(A)	When rotation is clockwise with constant ω	(P)	1
(B)	When rotation is clock wise with decreasing ω	(Q)	2
(C)	When rotation is clockwise with increasing ω	(R)	3
(D)	Just after clockwise rotation begins from rest	(S)	4

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q5. A block of mass m = 20 kg is kept at a distance R = 1m from central axis of rotation of a round turn table (A table whose surface can rotate about central axis). Table starts from rest and rotates with constant angular acceleration, α = 3 rad/sec^2. The friction coefficient between block and table is $\mu = 0.5$. At time $t = \frac{x}{3}$ sec from starting of motion (i.e. t = 0 sec) the block is just about to slip. Find the value of x.

SUBJECTIVE TYPE

Q6. A stone is thrown horizontally with the velocity 15m/s. Determine the tangential and normal accelerations of the stone in 1 second after it begins to move.

Q7. A stone is launched upward at 45° with speed v_0. A bee follows the trajectory of the stone at a constant speed equal to the initial speed of the stone.

(i) Find the radius of curvature at the top point of the trajectory.

(ii) What is the acceleration of the bee at the top point of the trajectory? For the stone, neglect the air resistance.

Q8. A thin circular loop of radius R rotates about its vertical diameter with an angular frequency ω. Show that a small bead on the wire loop remains at its lowermost point for $\omega \le \sqrt{g/R}$. What is the angle made by the radius vector joining the centre to the bead with the vertical downward direction for $\omega = \sqrt{2g/R}$? Neglect friction.

Q9. An aircraft executes a horizontal loop at a speed of 720 km/h with its wings banked at 15°. What is the radius of the loop ?

Q10. A circular racetrack of radius 300 m is banked at an angle of 15°. If the coefficient of friction between the wheels of a race-car and the road is 0.2, what is the (a) optimum speed of the race-car to avoid wear and tear on its tyres, and (b) maximum permissible speed to avoid slipping ?

Answer Key

Ex-1											
1	2	3	4	5	6	7	8	9	10	11	12
A	A	C	C	C	C	A	A	C	C	A	A
13	14	15	16	17	18						
A	A	C	D	C	B						

Ex- 1A

1. A, C, D **2.** A, C **3.** A, C **4.** A **5.** B

6. A **7.** B **8.** (i) $N_F = mg$, (ii) $N_W = \frac{mv^2}{r}$.

9. $\sqrt{2g}$ rad/s **10.** $\frac{v}{2d}$

Ex-2									
1	2	3	4	5	6	7	8	9	10
C	B	C	A	A	B	A	C	A	B

Ex- 2A

1. A, C **2.** A, B, C **3.** B, D **4.** (A)-R; (B)-S; (C)-Q; (D)-P

5. 2 **6.** $a_t = \frac{20}{\sqrt{13}}$, $a_n = \frac{30}{\sqrt{13}}$ **7.** (i) $\frac{V_0^2}{2g}$, (ii) 2g

8. 60° **9.** 15 km **10.** (a) 28.3 m/s (b) $v_{max} = 38.4\ ms^{-1}$.

-: Important Notes & Formulas :-